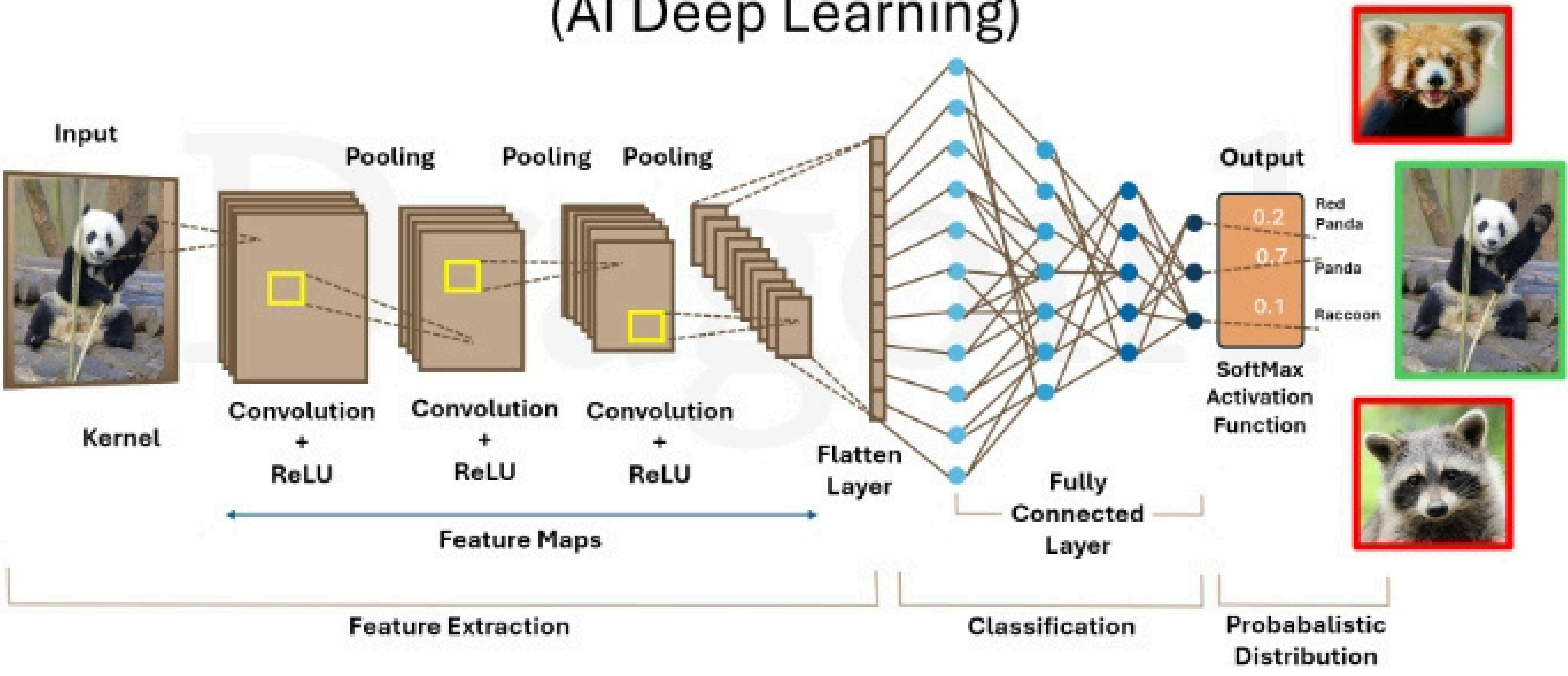
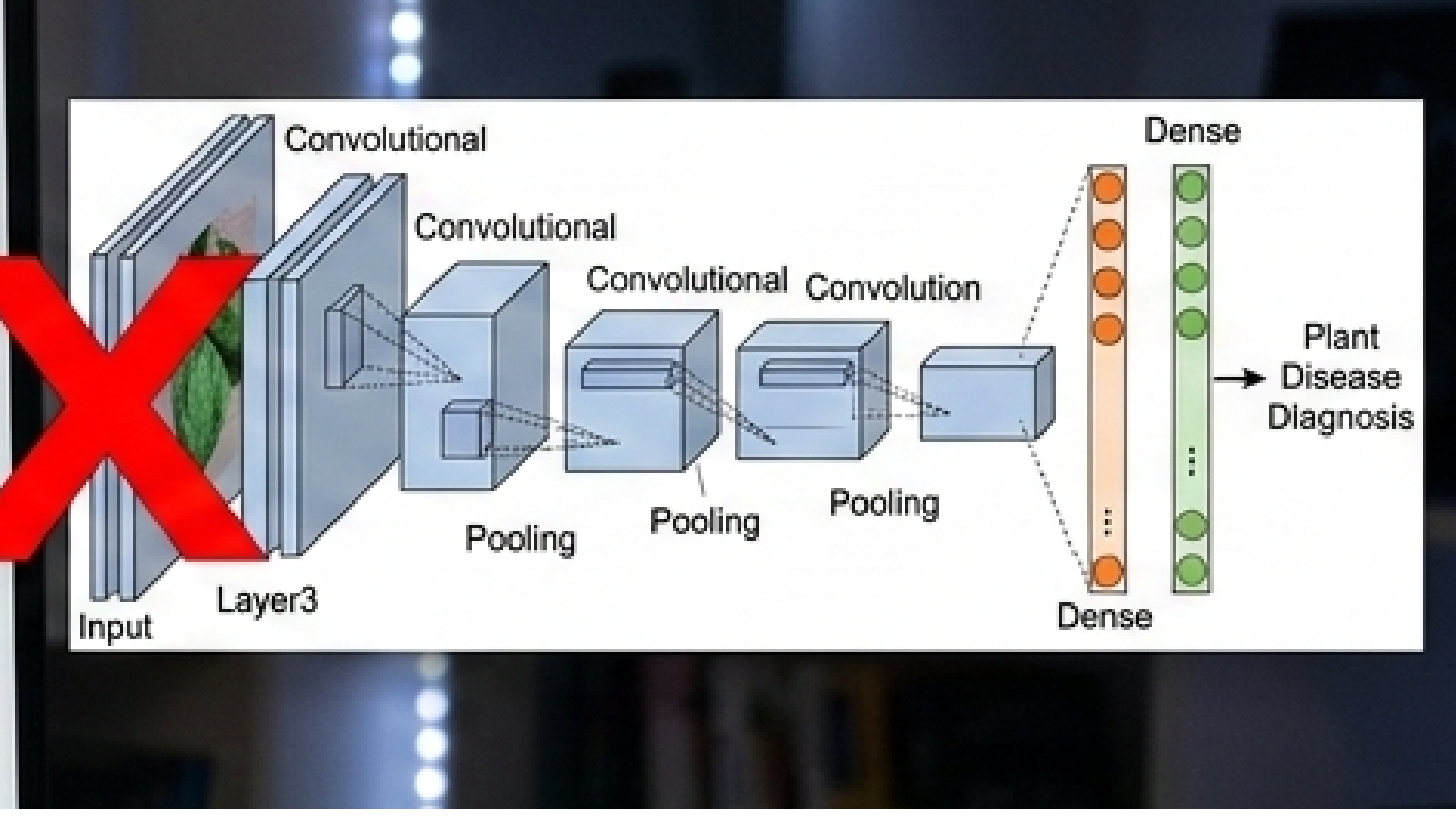
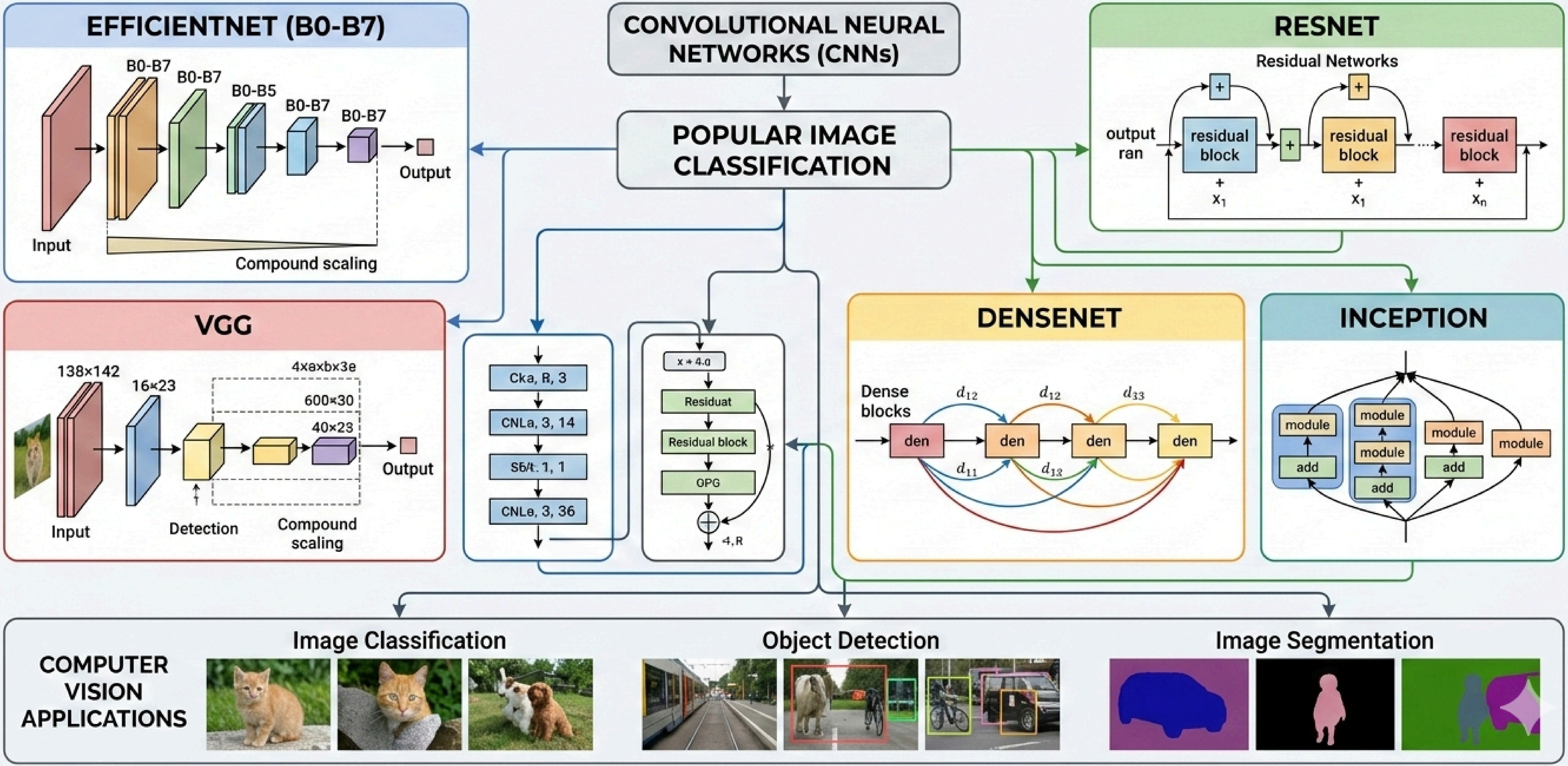
Convolutional Neural Networks

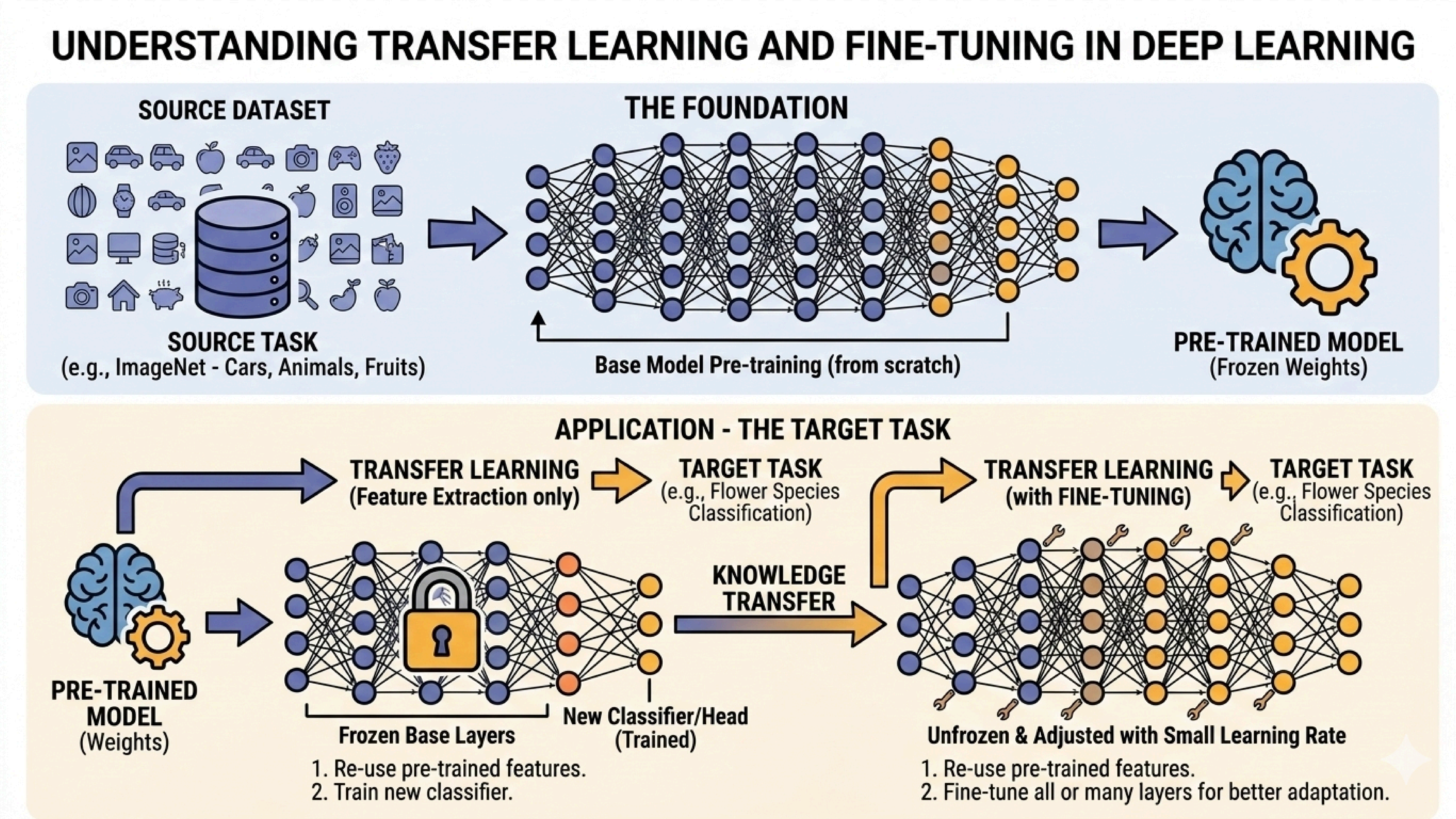
(AI Deep Learning)

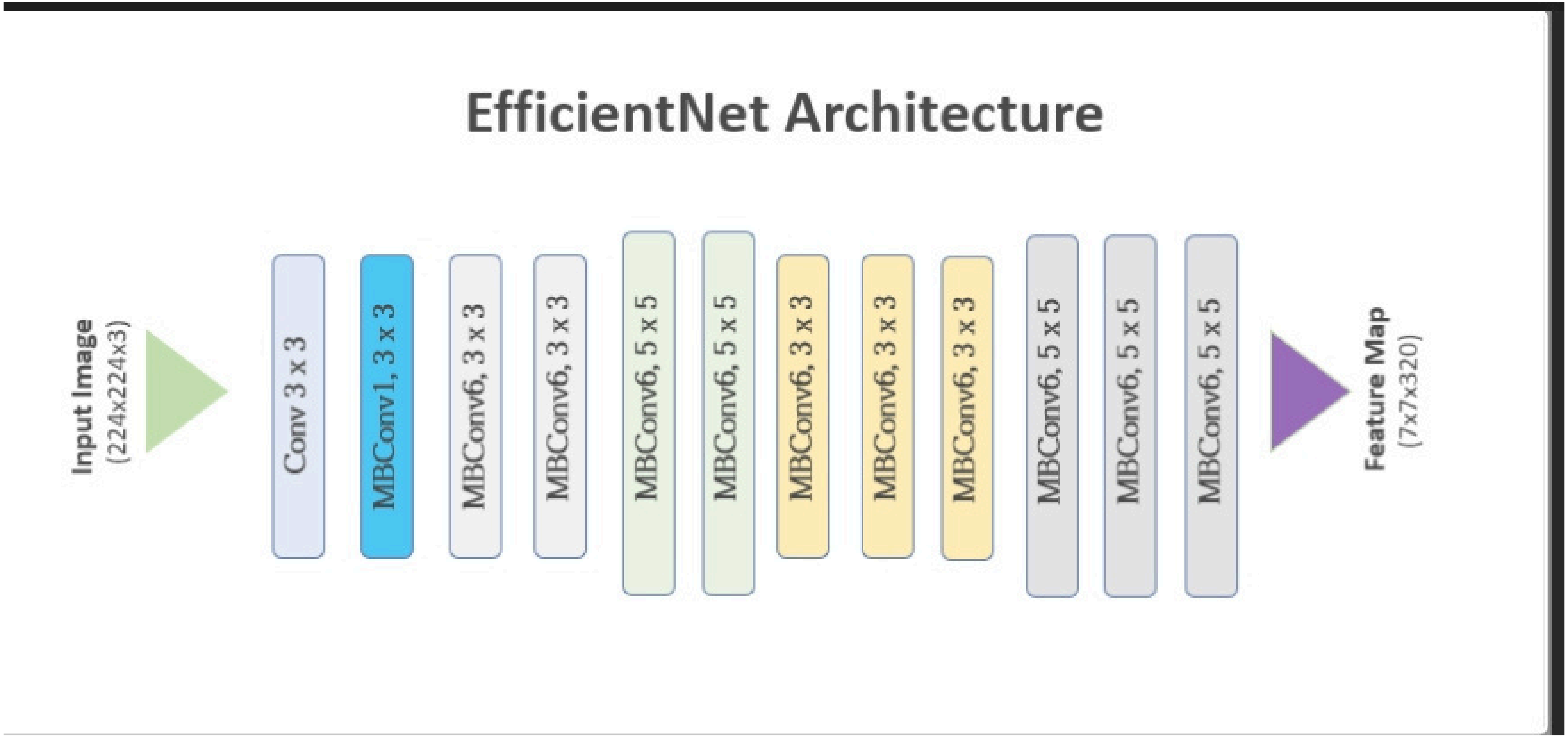


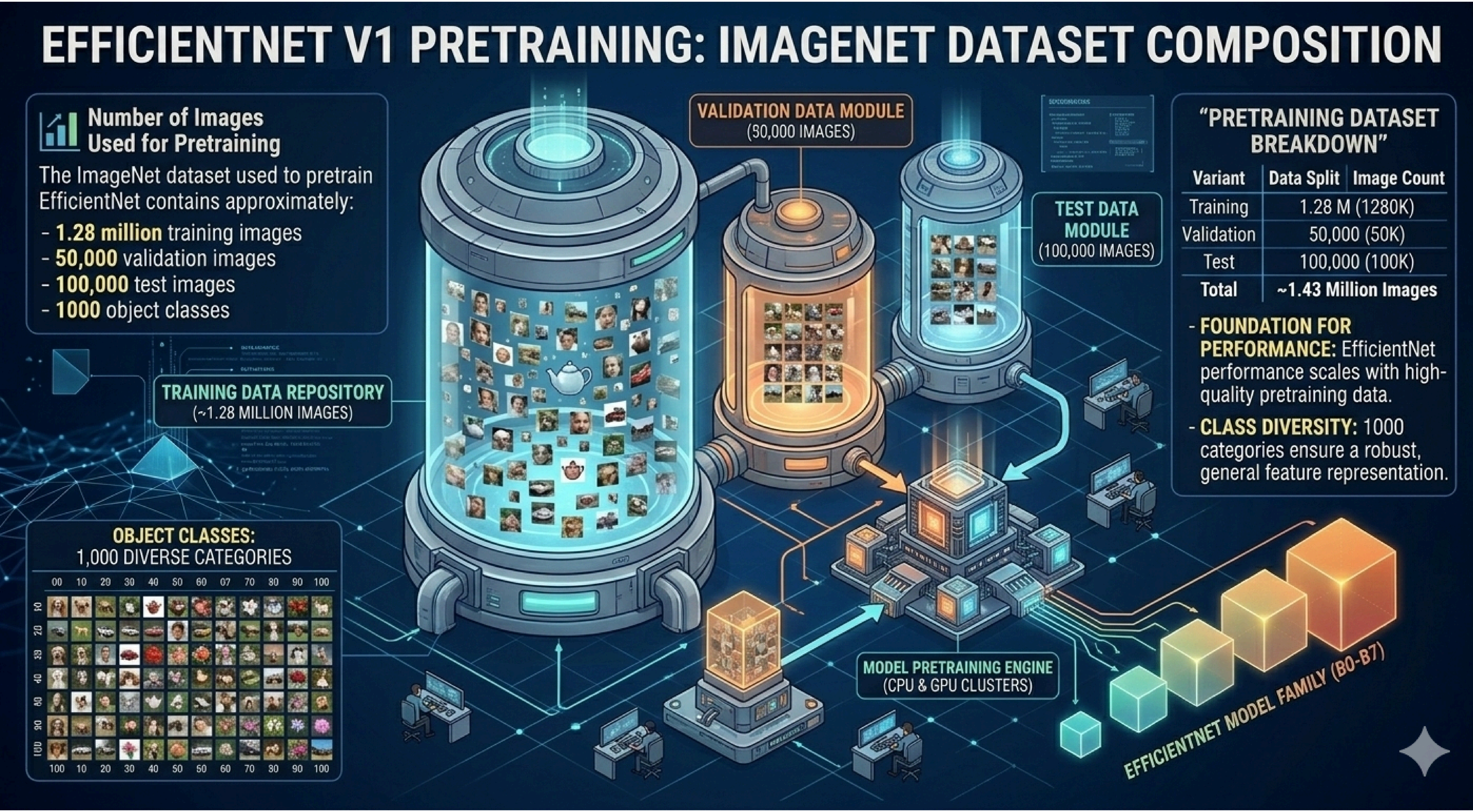

Convolutional
Convolutional
Convolutional
Convolution
Dense
Plant
Disease
Diagnosis
Pooling
Pooling
Pooling
Layer3
Input
Dense

LANDSCAPE OF CNN-BASED IMAGE MODELS




UNDERSTANDING TRANSFER LEARNING AND FINE-TUNING IN DEEP LEARNING
SOURCE DATASET
THE FOUNDATION
SOURCE TASK
(e.g., ImageNet - Cars, Animals, Fruits)
Base Model Pre-training (from scratch)
PRE-TRAINED MODEL
(Frozen Weights)
APPLICATION - THE TARGET TASK
TRANSFER LEARNING
(Feature Extraction only)
TARGET TASK
(e.g., Flower Species Classification)
TRANSFER LEARNING
(with FINE-TUNING)
TARGET TASK
(e.g., Flower Species Classification)
KNOWLEDGE TRANSFER
PRE-TRAINED MODEL
(Weights)
Frozen Base Layers
New Classifier/Head
(Trained)
1. Re-use pre-trained features.
2. Train new classifier.
Unfrozen & Adjusted with Small Learning Rate
1. Re-use pre-trained features.
2. Fine-tune all or many layers for better adaptation.

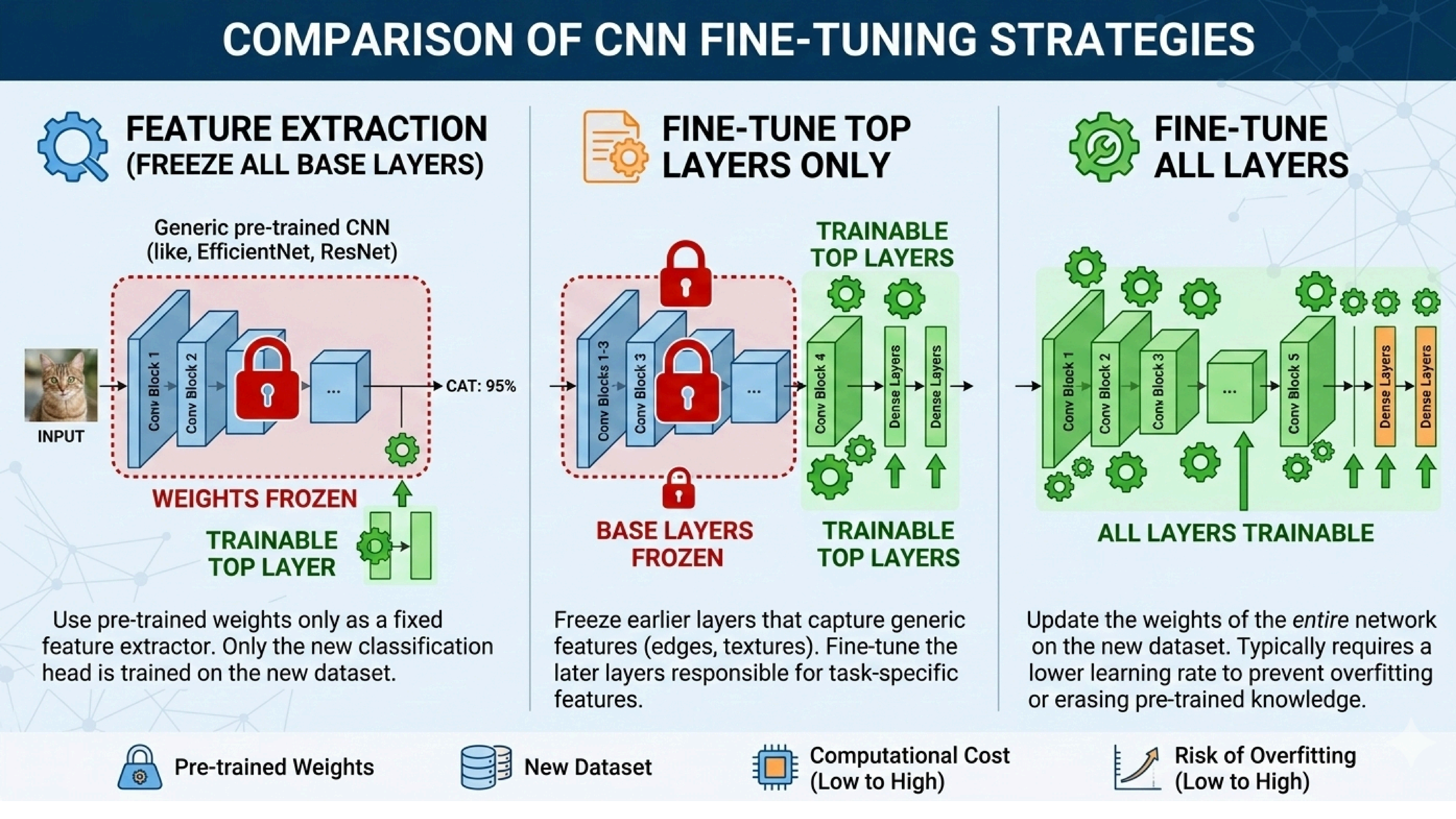

EfficientNet Architecture
Input Image
(224x224x3)
Conv 3 x 3
MBConv1, 3 x 3
MBConv6, 3 x 3
MBConv6, 3 x 3
MBConv6, 5 x 5
MBConv6, 5 x 5
MBConv6, 3 x 3
MBConv6, 3 x 3
MBConv6, 3 x 3
MBConv6, 5 x 5
MBConv6, 5 x 5
MBConv6, 5 x 5
Feature Map
(7x7x320)

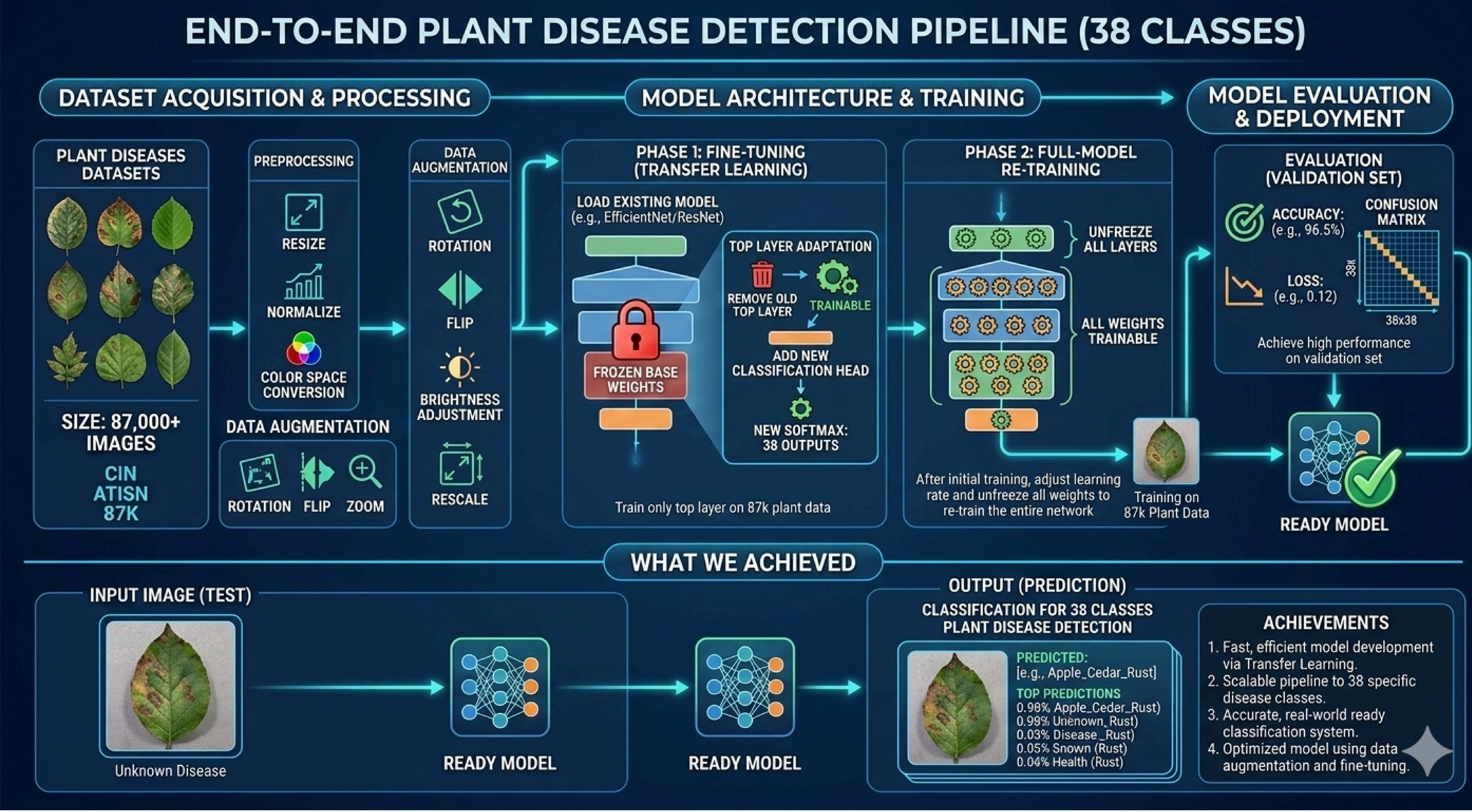


EFFICIENTNET V1 PRETRAINING: IMAGENET DATASET COMPOSITION
Number of Images Used for Pretraining
The ImageNet dataset used to pretrain EfficientNet contains approximately:
- 1.28 million training images
- 50,000 validation images
- 100,000 test images
- 1000 object classes
VALIDATION DATA MODULE (50,000 IMAGES)
TEST DATA MODULE (100,000 IMAGES)
TRAINING DATA REPOSITORY (~1.28 MILLION IMAGES)
"PRETRAINING DATASET BREAKDOWN"

Variant	Data Split	Image Count
Training		1.28 M (1280K)
Validation		50,000 (50K)
Test		100,000 (100K)
Total		~1.43 Million Images

- FOUNDATION FOR PERFORMANCE: EfficientNet performance scales with high-quality pretraining data.
- CLASS DIVERSITY: 1000 categories ensure a robust, general feature representation.
OBJECT CLASSES: 1,000 DIVERSE CATEGORIES
MODEL PRETRAINING ENGINE (CPU & GPU CLUSTERS)
EFFICIENTNET MODEL FAMILY (B0-B7)


COMPARISON OF CNN FINE-TUNING STRATEGIES
FEATURE EXTRACTION (FREEZE ALL BASE LAYERS)
Generic pre-trained CNN (like, EfficientNet, ResNet)
INPUT
Conv Block 1
Conv Block 2
...
CAT: 95%
WEIGHTS FROZEN
TRAINABLE TOP LAYER
Use pre-trained weights only as a fixed feature extractor. Only the new classification head is trained on the new dataset.
FINE-TUNE TOP LAYERS ONLY
TRAINABLE TOP LAYERS
Conv Blocks 1-3
Conv Block 3
...
Conv Block 4
Dense Layers
Dense Layers
BASE LAYERS FROZEN
TRAINABLE TOP LAYERS
Freeze earlier layers that capture generic features (edges, textures). Fine-tune the later layers responsible for task-specific features.
FINE-TUNE ALL LAYERS
Conv Block 1
Conv Block 2
Conv Block 3
...
Conv Block 5
Dense Layers
Dense Layers
ALL LAYERS TRAINABLE
Update the weights of the *entire* network on the new dataset. Typically requires a lower learning rate to prevent overfitting or erasing pre-trained knowledge.
Pre-trained Weights
New Dataset
Computational Cost (Low to High)
Risk of Overfitting (Low to High)


END-TO-END PLANT DISEASE DETECTION PIPELINE (38 CLASSES)
DATASET ACQUISITION & PROCESSING
MODEL ARCHITECTURE & TRAINING
MODEL EVALUATION & DEPLOYMENT
PLANT DISEASES DATASETS
SIZE: 87,000+ IMAGES
CIN
ATISN
87K
PREPROCESSING
RESIZE
NORMALIZE
COLOR SPACE CONVERSION
DATA AUGMENTATION
ROTATION
FLIP
ZOOM
DATA AUGMENTATION
ROTATION
FLIP
BRIGHTNESS ADJUSTMENT
RESCALE
PHASE 1: FINE-TUNING (TRANSFER LEARNING)
LOAD EXISTING MODEL (e.g., EfficientNet/ResNet)
FROZEN BASE WEIGHTS
TOP LAYER ADAPTATION
REMOVE OLD TOP LAYER
TRAINABLE
ADD NEW CLASSIFICATION HEAD
NEW SOFTMAX: 38 OUTPUTS
Train only top layer on 87k plant data
PHASE 2: FULL-MODEL RE-TRAINING
UNFREEZE ALL LAYERS
ALL WEIGHTS TRAINABLE
After initial training, adjust learning rate and unfreeze all weights to re-train the entire network
Training on 87k Plant Data
EVALUATION (VALIDATION SET)
ACCURACY: (e.g., 96.5%)
LOSS: (e.g., 0.12)
CONFUSION MATRIX
38x38
Achieve high performance on validation set
READY MODEL
WHAT WE ACHIEVED
INPUT IMAGE (TEST)
Unknown Disease
READY MODEL
READY MODEL
OUTPUT (PREDICTION)
CLASSIFICATION FOR 38 CLASSES PLANT DISEASE DETECTION
PREDICTED:
[e.g., Apple_Cedar_Rust]
TOP PREDICTIONS
0.98% Apple_Ceder_Rust)
0.99% Unenown_Rust)
0.03% Disease_Rust)
0.05% Snown (Rust)
0.04% Health (Rust)
ACHIEVEMENTS
1. Fast, efficient model development via Transfer Learning.
2. Scalable pipeline to 38 specific disease classes.
3. Accurate, real-world ready classification system.
4. Optimized model using data augmentation and fine-tuning.